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Meta-analyses

The effectiveness and acceptability of Mediterranean diet and calorie restriction in non-alcoholic fatty liver disease (NAFLD): A systematic review and meta-analysis

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SUMMARY

Background & aims: Lifestyle interventions that focus on reduced energy intake and improved dietary pattern are the mainstay of non-alcoholic fatty liver disease (NAFLD) management. However, it remains unclear which dietary approaches are most beneficial and promote greatest adherence. We aimed to synthesise data from randomised and clinical controlled trials, describing the effects of Mediterranean Diet and Calorie Restriction interventions on NAFLD surrogate markers, in adults.
Methods: We searched MEDLINE, Embase, Scopus, Web of Science, and Cochrane Central Register of Controlled Trials (October 2021). Study quality was assessed using the Cochrane Collaboration's tools: risk of bias for randomised controlled trials, and risk of bias in non-randomised studies of interventions. Meta-analyses were performed using a random effects model, and the I^2 statistic was used to assess heterogeneity.
Results: Of 4041 records identified, 26 articles with 3037 participants met the inclusion criteria, including studies on calorie-restricted interventions (CRI) (n 9), Mediterranean diet (MD) interventions (n 13) and MD component interventions (n 4). Studies were heterogeneous regarding intervention components, assessment of liver status and diet outcomes. 3 studies reported zero attrition and mean attrition rate for the remaining 23 studies was 14%. Post-intervention meta-analyses revealed that dietary interventions reduced alanine aminotransferase (ALT) ($P < 0.001$), aspartate aminotransferase (AST) ($P = 0.004$), Fatty Liver Index (FLI) ($P < 0.001$), hepatic steatosis (HS) ($P = 0.02$), and liver stiffness ($P = 0.01$). CRI had favourable effects on ALT ($P < 0.001$), HS ($P < 0.001$) and liver stiffness ($P = 0.009$). MD reduced ALT ($P = 0.02$), FLI ($P < 0.001$) and liver stiffness ($P = 0.05$). There was a dose–response relationship between degree of calorie restriction and beneficial effects on liver function and weight loss, suggesting that this approach should remain the cornerstone of NAFLD management. In addition, diet composition changes have potential for improving NAFLD and the limited data suggest that MD may be an effective diet therapy.
Conclusion: These results support the current guidelines in NAFLD. However, further studies, which robustly evaluate the effects of interventions on dietary intake, acceptability and sustainability of the interventions, and quality of life and other patient-related outcomes are needed to support effective care delivery.
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1. Introduction

Non-alcoholic fatty liver disease (NAFLD) affects 25% of the global population, driving chronic liver disease and consequent healthcare resource utilisation [1,2]. The rising burden of NAFLD parallels the increasing prevalence of obesity [3]. NAFLD is a spectrum of progressive liver disease, comprising steatosis (NAFL), steatohepatitis (NASH), fibrosis/cirrhosis and hepatocellular carcinoma (HCC) [4]. NAFLD is strongly associated with metabolic syndrome (MetS) and is an independent risk factor for development of cardiovascular disease (CVD) [4]. Indeed, longitudinal studies suggest that CVD-related mortality is more common than liver-related mortality among patients with NAFLD [5–7]. NAFLD management therefore requires clinicians to address both hepatic and extrahepatic manifestations of the MetS [3,8].

The prevalence of advanced liver disease and the demand for liver transplantation are growing [2,4]. Although there have been some promising advances, currently there are no regulatory approved pharmacological therapies [9,10]. Consequently, lifestyle interventions that encourage calorie (energy) restriction to induce weight loss and disease regression are the cornerstone of NAFLD management [8,11]. Changing dietary composition is also effective for CVD and Type 2 Diabetes (T2DM) risk reduction in patients with NAFLD [12]. However, despite the evidence that weight loss is advantageous, patients frequently find it difficult to achieve recommended targets [3]. Further, there is evidence that changes in diet composition are important in achieving beneficial hepatic effects, i.e., calorie restriction alone may not be the optimum treatment for NAFLD [8,12–14]. Thus, the need for optimum lifestyle-based strategies to encourage sustained weight loss remains unmet and many patients and clinicians are uncertain about which dietary approaches should be followed [15,16].

Recent research has shown that greater adherence to the Mediterranean Diet (MD) is effective for NAFLD prevention and management [8,17] and those randomised to a MD intervention had reduced CVD [18] but the precise dietary pattern associated with these benefits is unclear [17]. Definitive data are needed on whether a MD results in greater clinical benefits in NAFLD than other dietary interventions. Equally important, more evidence is needed on what dietary patterns are achievable by most patients and which promote greatest adherence. These knowledge gaps limit the delivery of effective diet and lifestyle interventions. We aimed to synthesise findings from randomised, and clinical controlled trials (RCTs/CCTs), investigating the effects of MD and calorie-restricted interventions (CRI), on markers of liver function, weight loss and on the acceptability of the intervention in patients with NAFLD.

2. Material and methods

2.1. Eligibility criteria

This systematic review was conducted according to Cochrane, the Centre for Reviews and Dissemination, and PRISMA guidelines [19–21]. The protocol has been registered with PROSPERO, the International Prospective Register of Systematic Reviews (Registration number CRD42019118537).

Only full reports of dietary interventions on surrogate markers of NAFLD, in adults (>18 years) were considered. Studies were selected according to the following criteria: a) adult participants (>18 years old) with NAFLD; b) RCTs and CCTs; c) MD interventions (that attempted to increase overall MD adherence, or which altered intake of its constituent food components), and CRI; d) NAFLD surrogate markers (i.e., hepatic steatosis (HS), NASH, hepatic fibrosis (HF), histological biomarkers, alanine aminotransferase (ALT) and aspartate aminotransferase (AST)).

The following were exclusion criteria: a) laboratory-based feeding trials; b) studies designed to test effects of specific micronutrients or dietary supplements; c) animal studies; d) in vitro studies; e) other study designs; f) studies testing specific exercise protocols. Studies testing dietary additives or other putative therapies were included only if MD effects were reported independently.

Primary outcomes included measures of NAFLD surrogate markers (i.e., HS, NASH, HF, histological biomarkers, ALT and AST). Secondary outcomes included body weight, waist circumference, cardiometabolic parameters, quality of life measures and intervention acceptability (i.e., attrition rates, where these data were the only measure of acceptability reported). Diet intake modification included changes in consumption of one or more dietary components and/or improved diet adherence at follow-up, assessed by diet assessment scores, quantitative analysis of self-reported food intake data (e.g., diet record diary) and biomarkers of dietary intake.

2.2. Search strategy and study selection

The following five electronic databases were searched systematically (October 2021): MEDLINE, Embase, Scopus, Web of Science, and Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library to identify eligible full text studies. A medical librarian (LE) assisted with the formulation of search strategies. The review of articles was restricted to those published in English with year of publication restricted to 2010–2021. An initial scoping search identified this as the period in which the relevant studies appeared. Reference lists of previously published related reviews were scanned to identify other studies eligible for inclusion. The search strategy involved combining words from the following concepts: NAFLD, controlled trials, MD and weight loss. Prior to searching, an extensive exercise to identify relevant terms was undertaken. The search terms were translated into a search strategy using a combination of index terms and keywords and refined iteratively. Highly sensitive search filters for identifying RCTs were used in MEDLINE and Embase [19,22]. The final list of search terms for all databases are provided (Supporting Information Tables 1–5).

2.3. Article selection and data extraction

Two reviewers (LH and CK or JG) independently assessed potentially relevant articles for eligibility. The decision to include or exclude studies was based initially on study title and abstract. Full text articles were then retrieved and evaluated to decide on final inclusion (with reasons for exclusion recorded). Where articles noted the existence of other study reports or published protocols, these were obtained to extract any potentially missing data. Data were extracted from the included studies using a standardised pre-piloted form (Supporting Information Table 6). Discrepancies between reviewers on article eligibility and data extraction were resolved through discussion with a third reviewer (JCM) using a consensus approach. We contacted study authors for additional data when required (Supporting Information Table 7).

Extracted information included: study design (country, recruitment methods, analysis type and methods, completion rates, intervention and comparator details), participant characteristics (population and setting, inclusion/exclusion criteria, baseline characteristics), and outcome assessment. These data were uploaded into Microsoft® Excel 2019 and used to compile a narrative synthesis of the results that is reported below using descriptive statistics (e.g., percentages) and summary tables.

2.4. Assessment of quality

Two reviewers (LH and CK or JG) independently assessed study quality using the Cochrane Collaboration's tools: risk of bias for RCTs (RoB 2) [23] and risk of bias in non-randomised studies of interventions (ROBINS-I tool) [24]. The data were then integrated into Microsoft® Excel and Access databases. Disagreements were discussed and resolved with a third reviewer (JCM). An overall summary score was calculated across domains using the defined criteria for each of the tools.

2.5. Meta-analysis and meta-regression

Meta-analysis was conducted using the Review Manager software (version 5.4, the Cochrane Collaboration, 2020), inputting mean and standard deviation (SD) data for post-intervention values. Data synthesis was undertaken, including the calculation of effects sizes with 95% CI, using a random-effects model with inverse variance weighting. Results of body weight, ALT, AST and Fatty Liver Index (FLI) were summarised as difference in means. However, due to the different techniques used to assess HS (histologic examination and magnetic resonance imaging (MR)) and liver stiffness (transient elastography (TE) Fibroscan™; ultrasound (USnd) acoustic radiation force impulse (ARFI), shear wave elastography (SWE) and Multiwave)), effects on these outcomes were summarised as standardised mean difference. Heterogeneity between studies was assessed by the χ^2 statistic (expressed as P value) and I^2 statistic (expressed as percentage) and significance was set at $p \leq 0.05$. For graphical presentation of the outcomes, forest plots were generated with studies arranged in order of effect size. Publication bias was assessed with funnel plots and by Egger's regression test.

Data not provided in the main text or tables of the original publications were extracted from the figures. Data from trials that reported their results as medians were converted to mean and SD using the formula described by Hozo et al., [25]. For trials with two intervention arms and a single control group, the total number of participants in each of the intervention arms were compared against half of the number of participants in the control group. A conservative method was adopted for crossover trials, with the mean and SD inputted separately for the intervention and control arms [26]. We conducted sensitivity analyses on the main meta-analysis models, to explore the robustness of the results by i) excluding crossover trials and ii) excluding trials that were judged as having serious, critical or high risk of bias [23,24]. We completed a subgroup analysis to explore the implications of different types of interventions; i) CRI, ii) MD interventions and iii) MD components interventions. Meta-regression analyses were used to determine whether baseline participant and study characteristics such as underlying NAFLD category, age, body mass index (BMI) and intervention duration, physical activity advice presence, sample size and attrition influenced the effect of dietary interventions on markers of liver function. Comprehensive meta-analysis (CMA) software version 3 (Biostat Inc.) was used to carry out meta-regression analysis using a random effects model.

3. Results

4041 publications were identified; from these 26 trials with a total of 3037 participants met the inclusion criteria. The summary reasons for exclusion of full text studies comprised ineligible population or intervention, insufficient detail on outcomes, and incorrect outcome measures, or analysis type (Supporting Information Table 8). Finally, all but five papers [27–31] were included in the meta-analysis. Results of the screening process are described in Fig. 1.

3.1. Study characteristics

The study characteristics are presented in Table 1. These included 54.7% male in the mixed-sex studies, with two studies including only males [32,33]. The mean (SD) age of participants was 48.3 (5.9) years and mean (SD) body mass index (BMI) 31.0 (2.3) kg/m^2 at baseline. The mean trial follow-up duration was 32.2 weeks (range 2–104 weeks). Study populations comprised NAFLD [27,31–50]; NASH [51,52]; individuals with abdominal obesity/dyslipidaemia [28,29] and those at risk of T2DM [30]. The trials were conducted in lower-middle income [41,43–45,50], upper-middle income [32,33,47,49,52] and high-income countries [27–31,34–40,42,46,48,51].

3.2. Eligible studies design and outcome measures

The study designs comprised twenty-three RCTs (88%) [28–39,41–47,49–52] and three CCTs (12%) [27,40,48]. There were two crossover trials (8%) [37,40] and the remaining trials used a parallel group design [27–36,38,39,41–52]. A broad range of reported outcomes and assessment methods were used. Nearly all studies assessed biochemical markers ALT (*n 25,* 96%) [27,28,30–52] and AST (*n 21, 81%*) [28,30–33,36,38–52]. Studies investigated HS using FLI (*n 6,* 23%) [27,32,33,36,38,39], imaging techniques (*n 20*, 77%) [28–34,36–39,41–49] and histology (*n 1*, 4%) [51]. HF was assessed by liver stiffness measures (LSM), using non-invasive techniques (*n 8,* 31%) [31,34,35,38,39,42,46,49] and scoring systems, NAFLD Fibrosis Score (NFS) (*n 3*, 12%) [27,32,35], Fibrosis-4 Index (FIB-4) (*n 1*, 4%) [27] and HepaScore (*n 1*, 4%) [34].

3.3. Intervention and comparison details

The included trials applied a diverse range of dietary interventions and comparator (control) treatments. Dietary interventions were delivered at various contact intervals, mainly face-to-face in individual consultations using nutrition counselling, and personalised approaches. Eight trials (31%) investigated multiple intervention arms (additional study arms from four trials were eligible for inclusion) [28,30,31,35]. Calorie reductions were applied in twenty studies (77%) [27,28,30–33,35,38–40,42,44–52], general physical activity was advised in fourteen studies (54%) [27–30,32,33,35,38,39,41,42,45,46,51] but only five studies (19%) reported specific behaviour change strategies [27,35,41,46,51]. The most common comparator (control) treatment was 'standard care', which was used in nineteen studies (73%) [27,28,30–32,35,36,38,39,41–44,46,47,49–52], followed by low-fat, high-carbohydrate diets (LFD) (*n 5*, 19%) [29,33,34,37,40]. The remaining studies applied a distinct dietary treatment or an individual dietary change component [45,48]. Seven studies (27%) provided participants with specific food items [28,29,34,37,47,48,51].

Across the twenty-six trials, nine investigated CRI (35%) [31,32,46–52], thirteen investigated MD interventions (50%) [27–29,33–42] and four investigated MD components (15%) [30,43–45] (Supporting Information Table 9). The most common interventions in the calorie-restricted trials involved broader, healthy dietary guidance (*n 5*, 19%) [32,46,50–52] or specific dietary treatments (*n 4*, 15%) [31,47–49]. The studies investigating MD interventions were mostly in Mediterranean regions/climates,

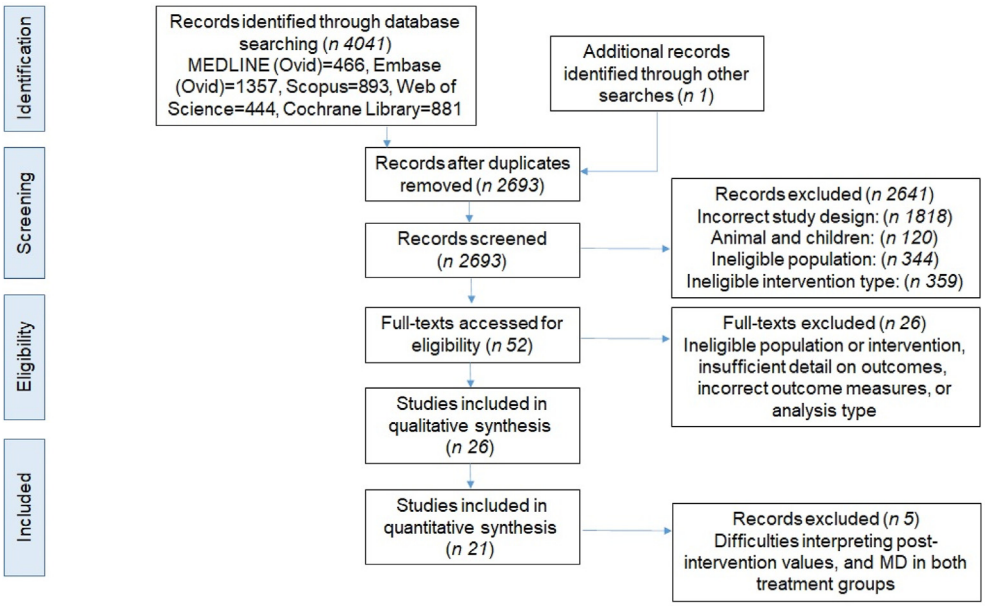

Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

and based on traditional Cretan diets, from descriptive food data and analysis of actual foods consumed [34,36,37], with low glycaemic index foods [36]; the MD pyramid, which adapts to different nutritional and socio-economic contexts of Mediterranean regions [35,53]; and national dietary guidelines [35,38]. Two trials adopted low-carbohydrate MD (<40 g/d for two months, then ≤70–80 g/d) supplemented with polyphenol-rich products [28,29], while a higher meal frequency (7 meals/d) was a key intervention feature of two trials [39,42]. The studies investigating MD components increased intakes of wholegrains [30] to at least half of cereal servings each day [43]; reduced red meat consumption (substituting it with turkey, fish, or chicken) [30]; and allocated either 20 g or 20% of the total fat (30% of total energy intake) to olive oil each day [44,45].

3.4. Assessment of risk of bias and publication bias

The risk of bias was assessed for all outcomes included in the review (Supporting Information Figures 1–23 and Tables 10–21). We have presented the risk of bias assessment for ALT as the most common primary outcome (Figs. 2 and 3 and Table 2).

The overall risk of bias is summarised in Supporting Information Tables 22–24 The majority of RCTs were judged overall as some concerns [28,29,32,35–37,42,45,47,49] or high risk of bias [30,33,34,39,41,43,44,50–52], with two RCTs judged at low risk of bias [31,46]. There was a high risk of bias in the domain of deviations from intended interventions for four RCTs [34,50–52] and six RCTs were judged at high risk of bias due to missing outcome data [30,33,39,41,43,44]. The CCTs were judged overall as moderate [27,48] or serious risk of bias [40]. There was either moderate [27,48] or serious risk of bias [40] due to confounding, moderate risk of bias due to missing data [27,40] and the selection of participants [27]. Investigation of potential publication bias was performed by producing funnel plots (Supporting Information Figures 24–29). Visual inspection of the funnel plots did not show evidence of publication bias for the outcomes described. Eggers's regression test confirmed the likely absence of publication bias in meta-analyses, but the power of the test was poor for three of the outcomes (FLI, LSM and HS) with <10 trials.

3.5. Effects of interventions on primary and secondary outcomes

Table 3 outlines the effects of dietary interventions on primary and secondary outcomes. The following sections consider CRI, MD interventions and MD component interventions separately.

3.5.1. Calorie-restricted interventions (CRI)

When compared with standard care (SC), CRI improved ALT [32,46,49], AST [52], FLI [32], and HS as measured by imaging (USnd/^{1}H-MRS) [31,32,46,47,49] and LSM [46,49]. These were associated with greater reductions in waist circumference [46] and body weight (5.8% (CRI) vs. 0.3% (SC)) [46,49–51]. CRI-induced body weight loss of ≥7% reduced waist circumference, and tended to improve ALT, HS, and histological NAFLD Activity Score (NAS) [51]. CRI improved Homeostatic Model Assessment of Insulin Resistance (HOMA-IR); fasting insulin [50,52]; as well as low-density lipoprotein (LDL) [46] and triglycerides (TG) [32]. CRI resulted in higher protein consumption [50] and, as planned, a fibre-enriched CRI increased fibre intakes [47].

Both a calorie-restricted diet and a carbohydrate-restricted diet improved AST and HS (^{1}H- NMR), but the carbohydrate-restricted diet reduced HS to a greater extent (12.0% vs. 5.0%), despite similar weight loss (~4.5%) [48]. Randomisation to a carbohydrate-restricted diet, resulted in higher protein and fat, and lower carbohydrate intakes [48].

3.5.2. MD interventions

When compared with standard care, MD improved ALT [28,35,41]; AST [28,41]; FLI [38]; HS assessed by imaging (^{1}H-MRS/MRI/USnd) [28,36,41,42]; LSM [35,38]; waist circumference [28,38], and resulted in larger body weight loss of 5.4% vs. 1.0% [28,35,38]. MD produced bigger reductions in HOMA-IR, fasting insulin, total cholesterol (TC) and TG [38]. MD adherence was higher in MD groups [35,39,42] with increased consumption of fibre [35], fruit [42] and nuts [28]. MD reduced calorie [39], carbohydrate [28,39,42], and sodium intake [42] alongside increased meal frequency, protein and polyunsaturated fat (PUFA) intake [39]. MD increased total plasma polyphenol and serum folate concentrations [28].

Table 1
Characteristics of the studies included in the review.

Author	Design	Male (n, %)	Clinical Group	Mean age (years)	Mean BMI (kg/m^2)	Intervention and Comparator (control) treatments	Intervention features	Duration (weeks)/contacts
Properzi 2018, Australia [34]	RCT	26, 51.0%	NAFLD	52.0	30.9	**Intervention:** MD Based on traditional Cretan diet (40% CHO, 20% PRO, 35–40% FAT and <10% SFA). **Comparator:** Low-fat/high CHO (LFD) Based on National Health and Medical Research Council and American Heart Association guidelines (50% CHO, 20% PRO, 30% FAT and <10% SFA).	**Diet food items:** 750 g nuts and 750 mL olive oil (MD) and 1 kg of natural muesli and 200 g of low-fat snack bars, every 4-wks (LFD).	12 Weekly calls (4-wks) and monthly contacts.
Katsagoni 2018, Greece [35]	RCT	43, 68.3%	NAFLD	48.3	31.2	**Intervention:** MD Based on MD pyramid and Greek guidelines, (45% CHO, 20% PRO and 35% FAT). **Comparator:** MDL as MD, with optimal sleep (≥7 and ≤ 9 h/d) and mid-day rest. **Control:** Standard care (healthy diet lifestyle)	**Cal restriction:** 1500 cal/d (women); 1800 cal/d (men) (all). **Physical activity**: Moderate-vigorous intensity 30 min/d/10,000 steps/d (MDL). **Behaviour change:** Goal setting (MD/MDL).	26 7 × 60min every 2-wks (2 m), then monthly (4 m) contacts.
Misciagna 2017, Italy [36]	RCT	72, 73.5%	NAFLD	–	–	**Intervention:** MD Based on traditional Cretan diet and low glycaemic index foods (≤10% SFA). **Control:** Standard care (healthy diet lifestyle) Based on INRAN guidelines.	Recommended diets provided in brochure format with no advice given on calorie allowances, physical activity or behaviour changes (both).	26 Monthly contacts.
Ryan 2013, Australia [37]	RCT crossover	6, 50.0%	NAFLD	55.0	32.0	**Intervention:** MD Based on traditional Cretan diet (40% CHO, 20% PRO and 40% FAT). **Comparator:** Low-fat/high CHO Based on National Health and Medical Research Council and American Heart Association guidelines (50% CHO, 20% PRO and 30% FAT).	**Diet food items:** Pre-cooked meals and ingredients, every 2-wks (MD/LFD).	18 Fortnightly contacts.
Abenavoli 2017, Italy [38]	RCT	18, 60.0%	NAFLD	42.6	30.0	**Intervention:** MD Italian RDAs, (50–60%, CHO, 15–20% PRO 50% veg PRO, <30% MUFA and PUFA, ≤10%, SFA, <300 mg cholesterol and 25–30 g fibre). **Control:** Standard care (habitual diet)	**Cal restriction:** 1400–1600 cal/d (MD). **Physical activity:** General (MD).	26 Weekly/monthly calls.
Marin-Alejandre 2019, Spain [39]	RCT	51, 52.0%	NAFLD	50.2	33.5	**Intervention:** MD Higher meal frequency (7 meals/d) (40%–45% CHO, 25% PRO (primarily veg PRO) and 30%–35% FAT)). **Control:** Standard care (healthy diet lifestyle) Based on American Heart Association guidelines, 3–5 meals/d (50%–55% CHO, 15% PRO and 30% FAT).	**Cal restriction:** 30% Cal deficit (both). **Physical activity:** 10,000 steps/d (both).	26 –
Biolato 2019, Italy [40]	CCT crossover	18, 90.0%	NAFLD	42.7	30.3	**Intervention:** MD 40% CHO, 20% PRO and 40% FAT <10% SFA. **Comparator:** Low-fat/high CHO 62% CHO, 20% PRO and 18% FAT.	**Cal restriction:** 1400 cal/d (both).	48 Fortnightly contacts.

(*continued on next page*)

Table 1 (*continued*)

Author	Design	Male (n, %)	Clinical Group	Mean age (years)	Mean BMI (kg/m^2)	Intervention and Comparator (control) treatments	Intervention features	Duration (weeks)/contacts
Ristic-Medic 2021, Serbia [33]	RCT	27, 100%	NAFLD	33.7	30.3	**Intervention:** MD 50% CHO, 15% PRO and >30%. **Comparator:** Low-fat/high CHO 60% CHO, 15% PRO and <25% FAT.	**Cal restriction:** 30% or 600−800 cal/d deficit (both). **Physical activity:** 30 min/d (both).	12 Regular contacts/calls.
Nourian 2020, Iran [41]	RCT	22, 32.0%	NAFLD	48.8	32.2	**Intervention:** MD Intervention based on the health belief model (HBM). **Control:** Standard care	**Physical activity:** General (MD). **Behaviour change:** HBM (MD).	8 8 × 60 min/wk contacts.
Abbate 2021, Spain [42]	RCT	51, 60.0%	NAFLD	53.2	33.9	**Intervention:** MD Higher meal frequency (7 meals/d) (40−45% CHO, (low glycaemic index) 25% PRO (mainly veg PRO)) and 30%−35% FAT (mainly MUFA and PUFA). **Control:** Standard care (healthy diet lifestyle) Based on AASLD guidelines (45−65% CHO, 10−35% PRO and 30%−35% FAT).	**Cal restriction:** 25−30% deficit (both). **Physical activity:** 10,000 steps/d (both).	26 Fortnightly (6 m), then monthly.
Yaskolka Meir 2021, Israela [28]	RCT	259, 88.1%	Central obesity/ dyslipidaemia NAFLD (62%)	51.1	31.3	**Intervention:** MD (green) <40 g/d CHO (2 m) then 80 g/day, ~40% FAT (PUFA and MUFA) and 3 polyphenol-rich products. **Comparator:** MD MD as MD (green) with 1 polyphenol-rich product. **Control:** Standard care (healthy diet lifestyle)	**Diet food items:** 28 g/d walnuts (MD groups); 3−4 cups/d green tea and 100 g/d *Wolffia globosa* (Mankai strain) as green shake replacing dinner (MD green); lunch (all). **Cal restriction:** 1200−1400 cal/d (women); 1500−1800 cal/d (men) (MD groups). **Physical activity:** Aerobic/ resistance 45−60min/3-4 d/wk (all).	78 90 min/wk (1 m), monthly (5 m) then every other month (12 m) contacts, text messages/website.
Gepner 2018, Israela [29]	RCT	247, 88.8%	Central obesity/ dyslipidaemia NAFLD (53%)	47.8	30.8	**Intervention:** MD <40 g/d CHO (2 m) then ≤70 g/day with 1 polyphenol-rich product (from 3 m). **Comparator:** Low-fat/high CHO 30% FAT ≤10% SFA and ≤300 mg/d cholesterol/d.	**Diet food items:** 28 g/d walnuts (MD); lunch (all). **Physical activity:** 60 min/m educational workshop Aerobic/resistance 30−60min/3 d/wk (for those randomised after 6 m) (both).	78 90 min/wk (1 m), then monthly.
Mazzotti 2018, Italya [27]	CCT	383, 53.5%	NAFLD	50.6	33.5	**Intervention:** MD Web-based intervention reproduces group sessions with interactive games, tests and mail contacts. **Comparator:** MD Group-based multi-disciplinary intervention.	**Cal restriction:** General (both). **Physical activity:** General (both). **Behaviour change:** Motivational interviewing; stimulus control; and weight loss maintenance strategies (both).	104 5 × 120 min/wk contacts.
Willmann 2019, Germanya [30]	RCT	46, 25.8%	Risk of T2DM	42.0	31.1	**Intervention:** MD component Reduced red meat consumption. **Comparator:** MD component Increased wholegrain consumption (≥40 g/d). **Control:** Standard care (habitual diet)	**Cal restriction:** 400 cal/d deficit (all). **Physical activity:** ≥3 h/wk (all).	26 Monthly contacts.

Dorosti 2020, Iran [43]	RCT	39, 34.8%	NAFLD	42.8	32.2	**Intervention:** MD component Increased wholegrain intake (≥½ of cereal servings/d). **Control:** Standard care (healthy diet lifestyle) **Both groups:** Based on Dietary Guidelines for Americans.	Dietary advice and education information provided with no advice given on calorie allowances, physical activity or behaviour changes (both).	12 Weekly/monthly contacts.
Shidfar 2018, Iran [44]	RCT	26, 52.0%	NAFLD	45.9	29.8	**Intervention:** MD component Increased olive oil intake (20% of total fat) **Control:** Standard care (healthy diet lifestyle) **Both groups:** 50% CHO, 20% PRO and 30% FAT.	**Cal restriction:** Personalised Cal deficit (both). Olive oil dosage supplied (MD component).	13 –
Rezaei 2019, Iran [45]	RCT	29, 43.9%	NAFLD	43.6	30.1	**Intervention:** MD component Increased olive oil intake (20 g/d). **Comparator:** Increased sunflower oil intake (20 g/d). **Both groups:** 50–55% CHO, 10–15% PRO and 30–35% FAT.	**Cal restriction:** 500 cal/d deficit (both). **Physical activity:** Moderate intensity 30–40 min/d (both). Olive oil and sunflower oil dosages provided (both).	12 Fortnightly/monthly contacts.
Wong 2013, Hong Kong [46]	RCT	72, 46.8%	NAFLD	51.0	25.4	**Intervention:** Cal-restricted dietary intervention Based on American Dietetic Association guidelines. **Control:** Standard care (healthy diet lifestyle)	**Cal restriction:** General (Cal-restricted). **Physical activity:** Moderate intensity/resistance training 30 min/d (Cal-restricted). 30 min/3 d/wk (control). **Behaviour change:** Coping with risky situations (Cal-restricted).	52 Weekly (4 m), monthly (8 m), and every three-month contacts.
Dong 2016, China [32]	RCT	280, 100%	NAFLD	57.3	25.8	**Intervention:** Cal-restricted dietary intervention 50–60% CHO, 15–20% PRO, 23–30% FAT. **Control:** Standard care (habitual diet)	**Cal restriction:** BMI dependent Cal balance (25–35 cal/kg/d) (Cal-restricted). **Physical activity:** Moderate-vigorous 30–60min/3-4 d/wk (Cal-restricted).	104 Three-month calls/annual contacts.
Promrat 2010, USA [51]	RCT	22, 71.0%	NASH	48.3	33.7	**Intervention:** Cal-restricted dietary intervention Based on American Heart Association; American Diabetic Association; American College of Sports Medicine, and Food Guide Pyramid. **Control:** Standard care (healthy diet lifestyle)	**Diet food items:** Commercial portion-controlled foods (Cal-restricted). **Cal restriction:** 1000–1200 cal/d < 91 kg BW; 1200-1500 cal/d > 91 kg BW (Cal-restricted). **Physical activity:** Moderate intensity 200 min/wk (Cal-restricted). **Behaviour change:** Stimulus control, problem solving and relapse prevention (Cal-restricted).	48 Weekly (6 m)/fortnightly (6 m) contacts.

(*continued on next page*)

Table 1 (*continued*)

Author	Design	Male (n, %)	Clinical Group	Mean age (years)	Mean BMI (kg/m^2)	Intervention and Comparator (control) treatments	Intervention features	Duration (weeks)/contacts
Cheng 2017, China [47]	RCT	13, 22.8%	NAFLD	60.0	26.9	**Intervention:** Cal-restricted dietary intervention Fibre-enriched diet. **Control:** Standard care (habitual diet)	**Diet food items:** Lunch as 30–40% of total energy intake/d (37–40% CHO, 9–13 g fibre, 5 g soluble fibre, 25–27% PRO, 35–37% FAT) (Cal-restricted). **Cal restriction:** General (Cal-restricted).	37 –
Browning 2011, USA [48]	CCT	5, 27.8%	NAFLD	44.5	35.0	**Intervention:** Low carbohydrate (<20 g d). **Comparator:** Cal-restricted dietary intervention	**Diet food items:** Meals (7–14/d) (both). **Cal restriction:** ~1200 cal/d (women) and ~1500 cal/d (men) (Cal-restricted).	2 –
Johari 2019, Malaysia [49]	RCT	33, 76.7%	NAFLD	49.0	29.9	**Intervention:** Cal-restricted dietary intervention Modified Alternate-Day Cal Restriction (MACR). **Control:** Standard care (habitual diet)	**Cal restriction:** 30% of Cal needs (fast days) and ad libitum (non-fast days) (Cal-restricted).	8 Intermittent calls/fortnightly contacts.
Shojasaadat 2019, Iran [50]	RCT	39, 51.3%	NAFLD	41.7	31.2	**Intervention:** Cal-restricted dietary intervention 52% CHO, 18% PRO and 30% FAT. **Control:** Standard care (habitual diet)	**Cal restriction:** 350–700 cal/d deficit (Cal-restricted).	12 –
Ghetti 2019, Brazil [52]	RCT	21, 47.7%	NASH	49.5	31.1	**Intervention:** Cal-restricted dietary intervention Based on American Dietetic Association guidelines and nutritional orientation (food guide Brazil). **Control:** Standard Care (healthy diet lifestyle) Nutritional orientation (food guide Brazil).	**Cal restriction:** 500–750 cal/d deficit for overweight/obese participants (Cal-restricted).	13 Monthly contacts.
Holmer 2021, Swedena [31]	RCT	33, 44.6%	NAFLD	56.3	32.4	**Intervention:** Cal-restricted dietary intervention (5:2). Based on Nordic Nutrition Recommendations on 5/d, (45–60% CHO, 10–20% PRO, 25% FAT). **Comparator:** Cal-restricted dietary intervention Low-carb high-fat (5–10% CHO, 15–40% PRO, 50–80% FAT). **Control:** Standard Care (healthy diet lifestyle)	**Cal restriction:** 500 cal/d (women) and 600 cal/d (men) on 2/d. 2000 cal/d (women) and 2400 cal/d (men) on 5/d (5:2). 1600 cal/d (women) and 1900 cal/d (men) (LFD).	12 Fortnightly calls (2,4 and 8wks) and 6wk contact.

USA, United States of America; AASLD, American Association for the Study of Liver Disease; CCT, controlled clinical trial; RCT, randomised controlled trial; NAFLD, non-alcoholic fatty liver disease; NASH, non-alcoholic steatohepatitis; T2DM, Type 2 Diabetes; MD, Mediterranean diet; MDL, Mediterranean diet lifestyle; LFD, low fat diet; HBM, health belief model; INRAN, Italian National Food Consumption Survey; RDAs, Recommended Dietary Allowances; cal, calorie; CHO, carbohydrate; PRO, protein; BW, body weight; BMI, body mass index; MUFA, monosaturated fat; PUFA, polyunsaturated fat; SFA, saturated fat; min, minute; h, hour; d, days; wks, weeks; m, months; mL, millilitre; g, gram; kg, kilogram; veg, vegetable.

[a] Study not included in meta-analyses.

Fig. 2. Risk of bias judgements of the included RCT studies for ALT based on reference [23]. * Study not included in meta-analyses. USA, United States of America.

Fig. 3. Risk of bias judgements of the included RCT crossover studies for ALT based on reference [23].

In comparison with LFD, MD induced either a comparable or greater improvement in ALT [33,34,40], AST [33], FLI [33] and HS as assessed by imaging (^{1}H-MRS/MRI) [29,34,37]. MD improved, high-density lipoprotein (HDL), TG [29,33] and waist circumference [29,40]. Both MD and LFD reduced body weight to similar degrees, except for one trial [40] in which weight loss was greater with MD (5.8% (MD) vs. 0.7% (LFD)) [40]. Participants on MD had higher levels of oleic acid, n-3 docosahexaenoic acid (DHA), monounsaturated (MUFA)/saturated fat (SFA) ratio and total MUFAs; and lower concentrations of palmitic acid and total SFAs in serum phospholipids [33]. MD resulted in higher intakes of fat, MUFA [34] and nuts [29] with lower intakes of carbohydrate [29,34], sugars, sodium [34], trans-fat and dietary cholesterol [29].

A web-based MD intervention and its group-based MD comparator both improved, ALT, FLI and FIB-4, but the web-based intervention normalised ALT more frequently, and produced bigger reductions in FLI [27].

Table 2
Risk of bias judgements of the included clinical controlled studies for ALT based on reference [24].

ROBINS-I	Confounding	Selection of participants into the study	Classifications of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Selection of the reported result	Risk of bias
Mazzotti 2018, Italy*								Moderate
Biolato 2019, Italy								Serious
Browning 2011, USA								Moderate

* Study not included in meta-analyses.
Green, low risk; yellow, moderate risk; orange, serious risk; USA, United States of America

3.5.3. MD component interventions

In comparison with standard care, increased consumption of wholegrains (≥½ of cereal servings/d) improved ALT, AST, fatty liver grades (USnd) and blood pressure (BP) [43]. Increased wholegrain consumption reduced calorie intake [43] and increased fibre intake [30], while a reduced red meat intervention lowered iron intakes and serum ferritin concentration [30]. Reduction in ferritin concentration correlated with improvement in liver fat content [30]. Supplemental olive oil (20 g or 20% of total fat/d) led to higher MUFA [44,45] and omega-3 intakes [45] and lower PUFA intakes [44,45], and produced greater improvements in ALT, AST [44] and fatty liver grades (USnd) [45].

3.6. Attrition

Three trials (12%) reported zero attrition [37,38,48]. For the remaining trials, attrition ranged from 2% to 45%, with a mean of 14%. Four trials (15%) reported attrition over 20% [27,30,39,47]. Where reported, reasons for dropouts included health problems or pregnancy [28,30,31,41,45–47,50], unwillingness to continue [28,29,32,41,42,45,47], non-adherence [31,32,34,40,45], scheduling conflicts/travel [45–47,49], personal reasons [28,29,31,34,47], dissatisfaction with either the allocated diet group, or diet phase [29,30,40], adoption of a non-intervention diet [41], alcohol excess [34], side effects [30], self-isolation due to COVID-19 [31] and death [32]. Studies investigating CRI reported the lowest attrition (mean 11%, range 2%–30%), followed by MD interventions (mean 16%, range 6%–45%), then MD components (mean 19%, range 14%–26%).

3.7. Meta-analysis and meta-regression of effects of dietary intervention on NAFLD surrogate markers

Twenty-one trials [32–52] were included in the meta-analysis (Supporting Information Table 25).

3.7.1. Primary outcomes

3.7.1.1. Effects of dietary interventions on ALT and AST. Meta-analysis showed that dietary interventions reduced post-intervention ALT (*n* 1295 participants) [32–35,37–52] (I^2: 67%, $P < 0.001$) and AST concentrations (*n* 1152 participants) [32,33,38–48,50–52] (I^2: 75%, $P = 0.004$) significantly, but there was substantial heterogeneity between the included trials. CRI (*n* 667 participants) [32,46–52] and MD interventions (*n* 418 participants) [33–35,37–42] had favourable effects on post-intervention ALT levels (I^2: 0%, $P < 0.001$), and (I^2: 81%, $P = 0.02$), respectively. There was no significant effect for MD component interventions (*n* 210 participants) [43–45] (I^2: 74%, $P = 0.09$) (Figs. 4 and 5). Meta-regression analyses found no significant moderating effects for underlying NAFLD category, age, BMI, intervention duration, physical activity advice presence, sample size and attrition.

3.7.1.2. Effects of dietary interventions on FLI and HS (by imaging and histology).. Dietary interventions (*n* 488 participants) [32,33,36,38,39] had favourable effects on post-intervention FLI values (I^2: 80%, $P < 0.001$), with substantial heterogeneity between the included trials. There were significant effects of MD interventions (*n* 228 participants) [33,36,38,39] on post-intervention FLI values (I^2: 71%, $P < 0.001$) (Fig. 6). Meta-analysis revealed a statistically significant effect of intervention (*n* 490 participants) [34,37,39,42,46–48,51] ($P = 0.02$) on post-intervention HS, but with substantial heterogeneity (I^2: 66%). Analysis by intervention type showed significant improvements in post-intervention HS following CRI (*n* 257 participants) [46–48,51] (I^2: 0%, $P < 0.001$), but no effect of MD (*n* 233 participants) [34,37,39,42] (I^2: 76%, $P = 0.70$). There were statistically significant subgroup differences (I^2: 73.9%, $P = 0.05$) (Fig. 7). Meta-regression analyses found no significant moderating effects for underlying NAFLD category, age, BMI, intervention duration, physical activity advice presence, sample size and attrition.

3.7.1.3. Effects of dietary interventions on LSM. Dietary interventions (*n* 464 participants) [34,35,38,39,42,46,49] lowered post-intervention liver stiffness, a surrogate for liver fibrosis (I^2: 81%, $P = 0.01$) but there was evidence of substantial heterogeneity. CRI (*n* 193 participants) [46,49] and MD interventions (*n* 271 participants) [34,35,38,39,42] had favourable effects on post-intervention liver stiffness (I^2: 0%, $P = 0.009$, and I^2: 87%, $P = 0.05$, respectively) (Fig. 8). Meta-regression analyses revealed that diet-induced improvement in liver stiffness decreased as participant age increased ($P < 0.001$) (Fig. 9). Although, there were no significant moderating effects for underlying NAFLD category, BMI, intervention duration, physical activity advice presence, sample size and attrition.

3.7.2. Secondary outcomes

3.7.2.1. Effects on dietary interventions on body weight. Post-intervention meta-analysis found no significant effect of dietary interventions (*n* 1226 participants) [32–35,37–40,42–52] on body weight (Fig. 10). Meta-regression analyses found no significant moderating effects for underlying NAFLD category, age, BMI, intervention duration, physical activity advice presence, sample size and attrition.

3.8. Sensitivity analyses

A sensitivity analyses, excluding results of two crossover trials (*n* 54 participants) [37,40] did not change the overall findings

Table 3
Effects of dietary interventions on primary and secondary outcomes.

Author	Outcome Measures	Results: primary outcomes (%)	Results: secondary outcomes (attrition rates n, %)	Results: secondary outcomes (cardiometabolics; anthropometry; QoL) (%)	Results: secondary outcomes (dietary intake)
Properzi 2018, Australia [34]	HS (MRS-PDFF), LSM (TE FibroScan™), HepaScore, ALT, cardiometabolics, anthropometry, QoL, and diet history.	↓ HS and ↓ ALT (both) HS RR: −32.4% ± 25.5% vs. −25.0% ± 25.3% (intervention vs. comparator) (ns)	**Intervention:** 2/26, 7.7% **Comparator:** 1/25, 4.0%	↓ GGT, HbA1c, TC, TG and FRS (intervention) ↓ WC and ↑ QoL (both) ↓ BW (both) BW: −2.3% vs. −2.1% (intervention vs. comparator) (ns)	↑ PRO, ↓ FAT, MUFA (comparator) ↓ CHO, sugars, and sodium; ↑ FAT and MUFA (intervention) ↓ SFA and ↑ fibre (both) ↑ FAT and MUFA; ↓ CHO, sugars, and sodium (intervention vs. comparator)
Katsagoni 2018, Greece [35]	LSM (USnd Multiwave), NFS, ALT, cardiometabolics, anthropometry, diet record diaries, FFQ and recalls.	↓ LSM (intervention/comparator vs. control) ↓ ALT (comparator vs. control)	**Intervention:** 0/21, 0.0% **Comparator:** 0/21, 0.0% **Control:** 7/21, 33.3%	↓ LDL and Non-HDL (intervention) ↓ Median BW (intervention: −5.4% and comparator: −6.3% vs. control: −2.1%)	↓ Cal (all) ↑ MUFA and PUFA (intervention) ↑ fibre and MD adherence (intervention/comparator vs. control). ↑ wholegrains, fruits, vegetables, and nuts; ↓ refined cereals, potatoes, red meat, and sweets (intervention/comparator)
Misciagna 2017, Italy [36]	HS (USnd), FLI, ALT, AST, cardiometabolics, anthropometry, diet record diaries and FFQ.	NAFLD score: negative interaction between time/intervention ↓ FLI and ALT (both)	**Intervention:** 6/50, 12.0% **Control:** 2/48, 4.2%	↓ GGT and HDL (intervention) ↓ TG and glycaemia (both)	–
Ryan 2013, Australia [37]	HS (^{1}H-MRS), ALT, cardiometabolics, anthropometry and diet record diaries.	↓ HS (intervention) HS RR: −39% ± 4% vs. −7%±3% (intervention vs. comparator)	**Intervention:** 0/12, 0.0% **Comparator:** 0/12, 0.0%	↓ HOMA-IR and fasting insulin (intervention) ↓ Systolic BP (both) BW: −1.1% vs. −2.6% (intervention vs. comparator) (ns)	↓ PRO, ↑ FAT, ↑ MUFA, fibre and ↓ SFA (intervention) ↑ CHO, PRO and ↓ FAT (comparator)
Abenavoli 2017, Italy [38]	HS (USnd), FLI, LSM (TE FibroScan™), ALT, AST, cardiometabolics and anthropometry.	↓ FLI and LSM (intervention) (intervention vs. control) ↑ AST (control)	**Intervention:** 0/20, 0.0% **Control:** 0/10, 0.0%	↓ TG, BW and WC (intervention) ↑ GGT, ↓ TC and LDL (both) ↑ fasting insulin and HOMA-IR (control) ↓ TC, TG, fasting insulin, HOMA-IR, BW and WC (intervention vs. control) Median BW: −6.0% vs. −0.5% (intervention vs. control)	–
Marin-Alejandre 2019, Spain [39]	HS (MRI), FLI, LSM (USnd ARFI), ALT, AST, cardiometabolics, anthropometry, diet questionnaire and FFQ.	↓ HS, FLI and ALT (both) ↓ AST (control) HS: −4.2% vs. −3.6% (intervention vs. control) (ns)	**Intervention:** 11/50, 22.0% **Control:** 11/48, 22.9%	↓ GGT, TG, FG, fasting insulin, HOMA-IR, and WC (both) ↓ Systolic and diastolic BP (both) ↓ BW (both) BW: −10.1% vs. −9.7% (intervention vs. control) (ns)	↓ Cal and SFA; ↑ meal frequency, PRO, fibre, and MD adherence (both) ↓ CHO and ↑ TAC and PUFA (intervention) ↓ Cal, CHO; ↑ meal frequency, PRO, PUFA and MD adherence (intervention vs. control)
Biolato 2019, Italy [40]	ALT, AST, cardiometabolics, anthropometry and diet questionnaire, FFQ and recalls.	↓ ALT and AST (intervention) ↓ ALT (intervention vs. comparator)	**Intervention:** 2/20, 10.0% **Comparator:** 2/14, 14.3%	↓ WC (intervention) (intervention vs. comparator) ↓ BW (intervention) BW: - 5.8% vs. −0.7% (intervention vs. comparator)	–
Ristic-Medic 2021, Serbia [33]	HS (USnd), FLI, ALT, AST, cardiometabolics, anthropometry, and FFQ.	↓ FLI, ALT and AST (both) ↓ FLI and AST (intervention vs. comparator)	**Intervention:** 2/14, 14.3% **Comparator:** 1/13, 7.7%	↓ GGT, TG, TC, LDL, FG, fasting insulin, HOMA-IR, BW, WC and ↑ HDL (both) ↓ TG and ↑ HDL (intervention vs. comparator) BW: −9.1% vs. −9.5% (intervention vs. comparator) (ns)	↓ n-6/n-3 ratio; ↑ n-3 PUFAs and DHA in serum and erythrocyte phospholipids; ↑ EPA in serum phospholipid (both). ↓ Palmitic acid and total SFAs; ↑ oleic acid, DHA, MUFA/SFA and

(*continued on next page*)

Table 3 (*continued*)

Author	Outcome Measures	Results: primary outcomes (%)	Results: secondary outcomes (attrition rates n, %)	Results: secondary outcomes (cardiometabolics; anthropometry; QoL) (%)	Results: secondary outcomes (dietary intake)
					MUFAs in serum phospholipids (intervention vs. comparator)
Nourian 2020, Iran [41]	HS (USnd), ALT, AST, anthropometry, HBM questionnaire, and diet record diaries.	↓ HS (both) (intervention vs. control) ↓ ALT (intervention) ↓ AST (both) ↓ ALT and AST (intervention vs. control)	**Intervention:** 5/41, 12.2% **Control:** 8/41, 19.5%	Improved knowledge and HBM variables (intervention) (intervention vs. control)	–
Abbate 2021, Spain [42]	HS (MRI), LSM, ALT, AST, cardiometabolics, anthropometry, FFQ and questionnaire.	↓ HS (both) (intervention vs. control) ↓ ALT (both) ↓ AST (control) HS: (intervention: −6.6% vs. control: −4.9%)	155 randomised; 128 analysed (17.4%); 85 included. **Intervention:** 43 **Control:** 42	↓ GGT, FG (control) ↓ TC and HbA1c (intervention) ↓ TG, fasting insulin, HOMA-IR, BP, serum ferritin and ↑ HDL (both) ↓ BW (both) ↓ WC (intervention) BW: (intervention: −8.0% vs. control: −6.2% (ns)	↑ MD adherence; ↓ CHO (both) (intervention vs. control) ↓ Cal, FAT, SFA, TFA, cholesterol, animal fat, meat/meat products; ↑ fibre and legumes (both) ↓ PRO, Vit B3, alcohol and sweets/ pastries (control) ↑ Mg, P, K, Vit B6, folic acid, milk/ dairy, fish, nuts, vegetable sources; ↓ cereals (intervention) ↓ Sodium; ↑ fruit (intervention) (intervention vs. control)
Yaskolka Meir 2021, Israela [28]	HS (^{1}H-MRS), ALT, AST, cardiometabolics, anthropometry and FFQ.	↓ HS (intervention vs. comparator/control) ↓ ALT (intervention/comparator) (intervention/comparator vs. control) ↓ AST (intervention/comparator) Median HS: (intervention: −2.0% vs. comparator: −1.1% and control: −0.7%)	**Intervention:** 9/98, 9.2% **Comparator:** 14/98, 14.3% **Control:** 7/98, 7.1%	↓ ALP (intervention/comparator) (intervention/comparator vs. control) ↓ BW (intervention/comparator) ↓ WC (all) (intervention vs. control) ↓ BW: (intervention: −4.0% and comparator: −2.9% vs. control: −0.4%)	↓ CHO and ↑ walnuts (intervention/ comparator vs. control); ↓ red meat and poultry; ↑ green tea and green shake (intervention vs. comparator); ↑ serum folate (all) (intervention vs. control); ↑ plasma polyphenols (intervention/ comparator vs. control)
Gepner 2018, Israela [29]	HS (MRI), cardiometabolics, anthropometry and FFQ.	↓ HS (intervention vs. comparator)	**Intervention:** 17/139, 12.2% **Comparator:** 21/139, 15.1%	↓ TG, diastolic BP and ↑ HDL (intervention vs. comparator) ↓ BW (both) ↓ WC (intervention vs. comparator)	↓ Cal, PRO and PUFA (both) ↓ CHO, TFA and cholesterol (both) (intervention vs. comparator) ↓ FAT, MUFA and SFA (both) (comparator vs. intervention) ↑ nuts (intervention vs. comparator)
Mazzotti 2018, Italya [27]	FLI, NFS, FIB-4, ALT, cardiometabolics, anthropometry, diet questionnaire.	↓ FLI, FIB-4 and ALT (both) ↓ FLI (intervention vs. comparator)	**Intervention:** 160/278, 57.6% **Comparator:** 137/438, 31.3%	↓ BW (both) BW: −5.5% vs.-4.2% (intervention vs. comparator)	↓ Cal (both)
Willmann 2019, Germanya [30]	HS (^{1}H-MRS), ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS (intervention/comparator) ↓ AST (comparator/control) HS RR: −29% vs. −26% vs. −10% (intervention vs. comparator vs. control) (ns)	**Intervention:** 10/58, 17.2% **Comparator:** 16/60, 26.7% **Control:** 20/60, 33.3%	↓ GGT, FG, TC and BW (all) ↓ Fasting insulin (intervention/ comparator) ↓TG (intervention/control) BW: −4.4% vs. −3.0% vs. −3.4% (intervention vs. comparator vs. control) (ns)	↓ Cal, CHO, PRO, and FAT (all) ↓ iron (intervention) ↑ iron (comparator); (intervention vs. comparator/control) ↑ fibre (comparator/control) (comparator vs. intervention/ control).
Dorosti 2020, Iran [43]	HS (USnd), ALT, AST, cardiometabolics, anthropometry and recalls.	↓ HS, ALT, and AST (intervention vs. control)	**Intervention:** 9/56, 16.1% **Control:** 9/56, 16.1%	↓ GGT, systolic and diastolic BP (intervention vs. control) ↓ BW -1.8% vs. −1.0% (intervention vs. control) (ns)	↓ Cal (intervention vs. control) ↑ fibre (intervention) ↑ wholegrains and FAT (intervention vs. control); ↑ grains, vegetables, and meat (control vs. intervention)
Shidfar 2018, Iran [44]	HS (USnd), ALT, AST, anthropometry, and diet questionnaire.	↓ HS and AST (intervention) ↓ALT (both) ↓ALT and AST (intervention vs. control)	**Intervention:** 4/25, 16.0% **Control:** 3/25, 12.0%	↓ WC (both) ↓ BW (both) BW: −4.3% vs.-3.5% (intervention vs. control) (ns)	↓ Cal, CHO, PRO, FAT, PUFA, and SFA (both). ↑ MUFA and ↓ PUFA (intervention vs. control)
Rezaei 2019, Iran [45]					

	HS (USnd), ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS and AST (both) ↓ ALT (comparator) ↓ HS (intervention vs. comparator)	**Intervention:** 6/32, 18.8% **Comparator:** 6/34, 17.6%	↓ TG (intervention) ↓ BW, WC, and systolic/diastolic BP (both) BW: −4.1% vs. −2.9% (intervention vs. comparator) (ns)	↑ MUFA, ↑ omega-3 and ↓ PUFA (intervention vs. control)
Wong 2013, Hong Kong [46]	HS (^{1}H-MRS), LSM (TE FibroScan™), ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS, LSM and ALT (intervention vs control) HS: −6.7% ± 6.1% vs. −2.1% ± 6.4% (intervention vs. control)	**Intervention:** 3/77, 3.9% **Control:** 6/77, 7.8%	↓ LDL, BW and WC (intervention vs. control) BW: −8.0% vs. −0.9% (intervention vs. control)	–
Dong 2016, China [32]	HS (USnd), FLI, NFS, ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS, ALT and FLI (intervention) (intervention vs. control) ↓ NFS (both)	**Intervention:** 11/141, 7.8% **Control:** 9/139, 6.5%	↓ TC, LDL and ↑ HDL (both) ↓ TG (intervention) (intervention vs. control) ↓ BW and WC (intervention) BW: −1.8% vs. +0.1% (intervention vs. control) (ns)	–
Promrat 2010, USA [51]	Histopathology, NAS, ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS, NAS, and ALT (intervention vs control) ↓ballooning injury and AST (both) HS: −1.1% ± 0.8% vs. −0.3% ± 0.8% (intervention vs control)	**Intervention:** 1/21, 4.8% **Control:** 0/10, 0.0%	↓ BW and WC (intervention vs. control) BW: −9.3% vs −0.2% (intervention vs. control)	–
Cheng 2017, China [47]	HS (^{1}H-MRS), ALT, AST, cardiometabolics and anthropometry.	↓HS (intervention) HS RR: −23.2% vs +20.9% (intervention vs. control)	**Intervention:** 6/28, 21.4% **Control:** 11/29, 37.9%	BW: −1.2% vs. +0.3% (intervention vs. control) (ns)	↑ fibre (intervention vs. control)
Browning 2011, USA [48]	HS (^{1}H NMR), ALT, AST, cardiometabolics, anthropometry and diet record diaries.	↓ HS and AST (both) HS: −12.0% vs. −5% (intervention vs. comparator)	**Intervention:** 0/9, 0.0% **Comparator:** 0/9, 0.0%	↓ TG and ↓ BW (both) BW: −4.7% vs. −4.2% (intervention vs. comparator) (ns)	↓ CHO; ↑ PRO and FAT (intervention vs. comparator)
Johari 2019, Malaysia [49]	HS (USnd), LSM (USnd SWE), ALT, AST, cardiometabolics, anthropometry, diet record diaries and recalls.	↓ HS, LSM, ALT and AST (intervention) ↓ HS, LSM and ALT (intervention vs. control)	**Intervention:** 3/33, 9.1% **Control:** 1/10, 10.0%	↓ FG (intervention) ↓ BW (intervention) BW: −2.5% vs. +1.1% (intervention vs. control)	–
Shojasaadat 2019, Iran [50]	ALT, AST, cardiometabolics, anthropometry, FFQ and recalls.	↓AST (intervention)	**Intervention:** 3/38, 7.9% **Control:** 4/38, 10.5%	↓ GGT, BW and WHR (both) ↓ FG, fasting insulin, HOMA-IR, TC, LDL and HDL (intervention) ↓ BW, fasting insulin, and HOMA-IR (intervention vs. control) BW: −3.4% vs. −1.2% (intervention vs. control)	↓ Cal, PUFA and fibre; ↑ PRO (intervention) ↑ PRO (intervention vs. control)
Ghetti 2019, Brazil [52]	ALT, AST, cardiometabolics, anthropometry and FFQ.	↓ALT and ↓AST (intervention) ↓AST (intervention vs. control)	**Intervention:** 2/22, 9.1% **Control:** 2/22, 9.1%	↓ GGT, FG, HOMA-IR, TC, TG, BW and WC (intervention) ↓ GGT, HOMA-IR, and fasting insulin (intervention vs. control) BW: −4.7% vs +0.7% (intervention vs. control) (ns)	–
Holmer 2021, Sweden[a] [31]	HS (MRS), LSM (TE FibroScan™), ALT, AST, cardiometabolics, anthropometry, diet record diaries and recalls	↓ HS (all) (intervention/comparator vs. control) ↓ LSM (intervention/control) (intervention/control vs. comparator) ↓ ALT (all) ↓ AST (control) HS: −6.1% and −7.2% vs. −3.6% (intervention and comparator vs. control)	**Intervention:** 1/25, 4.0% **Comparator:** 5/25, 20.0% **Control:** 4/24, 16.7%	↓TC and LDL (intervention) ↑ HDL ↓ BP (comparator) ↓ TG, HbA1c and HOMA-IR (intervention/comparator) ↓ BW, WHR (all) BW: intervention: −7.4% vs comparator: −7.7% vs. control: −2.6% (ns)	↓ Cal and SFA (intervention/control) ↓ MUFA and fibre (intervention) ↓ CHO, fibre; ↑ ALA and LA (plasma), FAT, SFA, PUFA and MUFA (comparator) ↑ PRO (intervention/comparator)

USA, United States of America; HS, hepatic steatosis; RR, relative reduction; USnd, ultrasound; LSM, liver stiffness measurement; ARFI, acoustic radiation force impulse; MRS-PDFF, magnetic resonance spectroscopy-measured proton density fat fraction; ^{1}H-MRS, proton magnetic resonance spectroscopy; MRI, magnetic resonance imaging; ^{1}H NMR, proton nuclear magnetic resonance; TE, transient elastography; SWE, shear wave elastography; FLI, Fatty Liver Index; NAFLD, non-alcoholic fatty liver disease; NFS, NAFLD Fibrosis Score; FIB-4, Fibrosis-4 Index; NAS, NAFLD Activity Score; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ALP, alkaline phosphatase; GGT, gamma-glutamyl transferase; QoL, quality of life; health belief model; BW, body weight; WC, weight circumference; BMI, body mass index; WHR, waist-to-hip ratio; CVR, cardiovascular risk; FG, fasting glucose; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; glycated haemoglobin, HbA1c; TC, total cholesterol; LDL, low-density lipoprotein; HDL, high-density lipoprotein; Non-HDL, non-high-density lipoprotein; TG, triglycerides; FRS, Framingham risk score; BP, blood pressure; FFQ, food frequency questionnaire; Cal, calorie; CHO, carbohydrate, PRO, protein; MUFA, monounsaturated fat; PUFA, polyunsaturated fat; SFA, saturated fat; TFA, trans-fat, n-3 PUFA, omega 3 polyunsaturated fatty acid; MD, Mediterranean diet, ALA; α-linolenic acid, LA; linoleic acid; DHA, n-3 docosahexaenoic acid; EPA, n-3 eicosapentaenoic acid; TAC, total antioxidant capacity; Mg, magnesium; K, potassium; P, phosphorus; Vit, vitamin; B3, Niacin; B6, pyridoxine; ns, not significant.

[a] Study not included in meta-analyses. HS and BW data presented as mean (unless indicated otherwise).

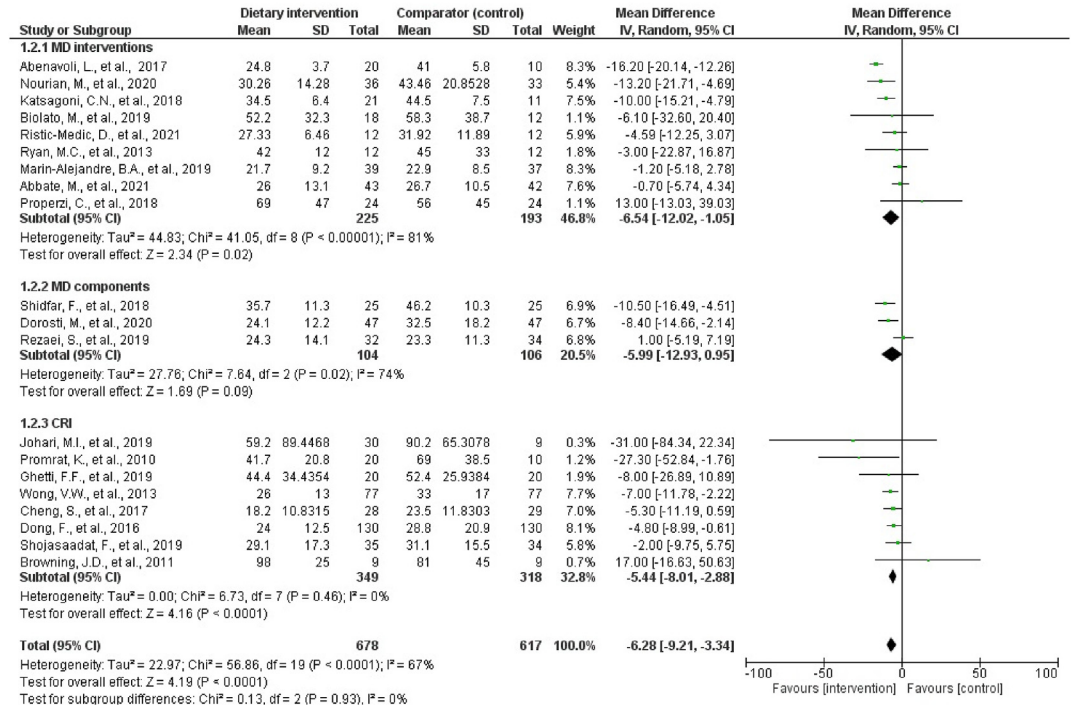

Fig. 4. Effects of dietary interventions on alanine aminotransferase (ALT). SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

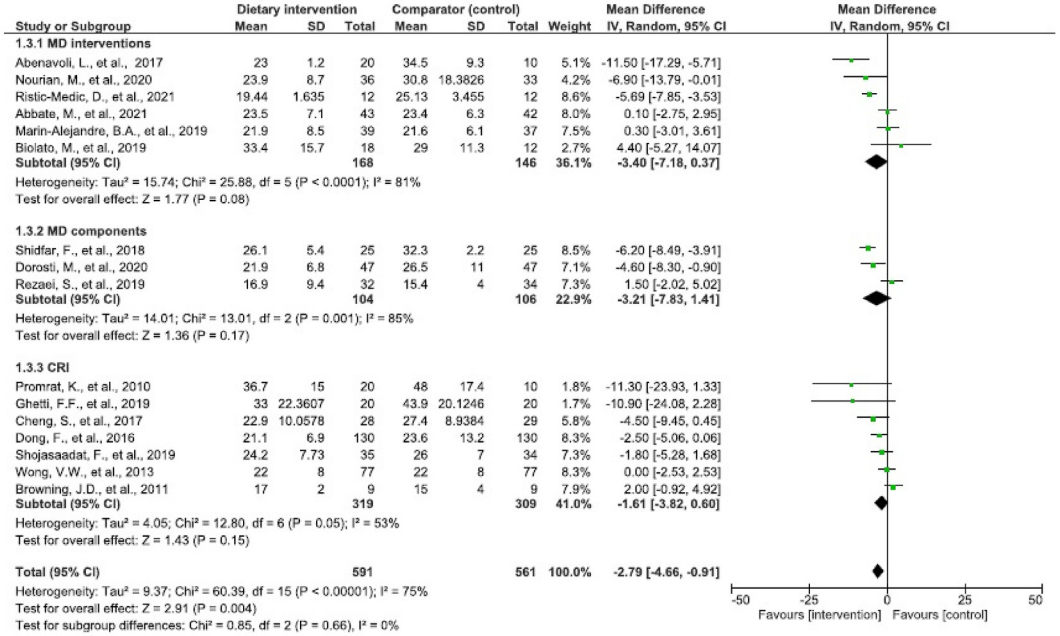

Fig. 5. Effects of dietary interventions on aspartate aminotransferase (AST). SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

(Supporting Information Table 26, Figures 30–33). Further sensitivity analyses were conducted, excluding results of ten trials (*n* 530 participants) that were judged as having serious [40], or high risk of bias [33,34,39,41,43,44,50–52]. After doing so, most of the findings were unaffected (Supporting Information Table 27, Figures 34–39). However, excluding these trials, revealed a significant effect of dietary interventions (*n* 765 participants) on post-intervention body weight (I^2: 26%, $P = 0.03$), and a significant effect of MD interventions (*n* 109 participants) on post-intervention HS (I^2: 0%, $P = 0.01$). There was no significant effect of dietary interventions (*n* 670 participants) on post-intervention AST (I^2: 74%, $P = 0.24$).

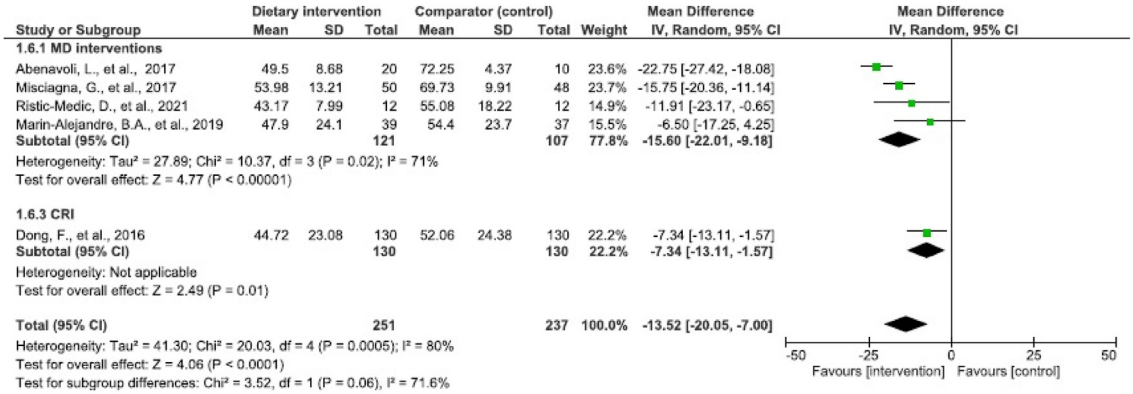

Fig. 6. Effects of dietary interventions on Fatty Liver Index (FLI). SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

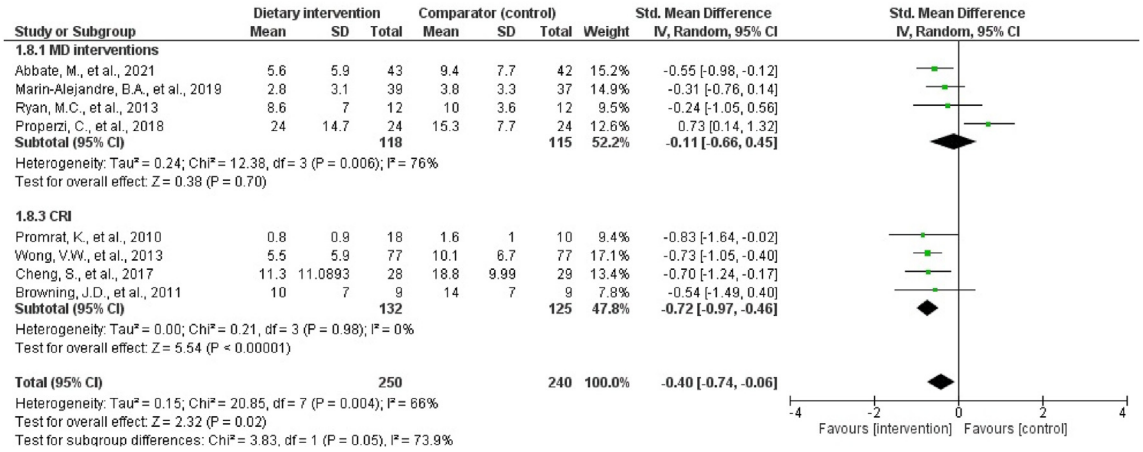

Fig. 7. Effects of dietary interventions on hepatic steatosis (HS). SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

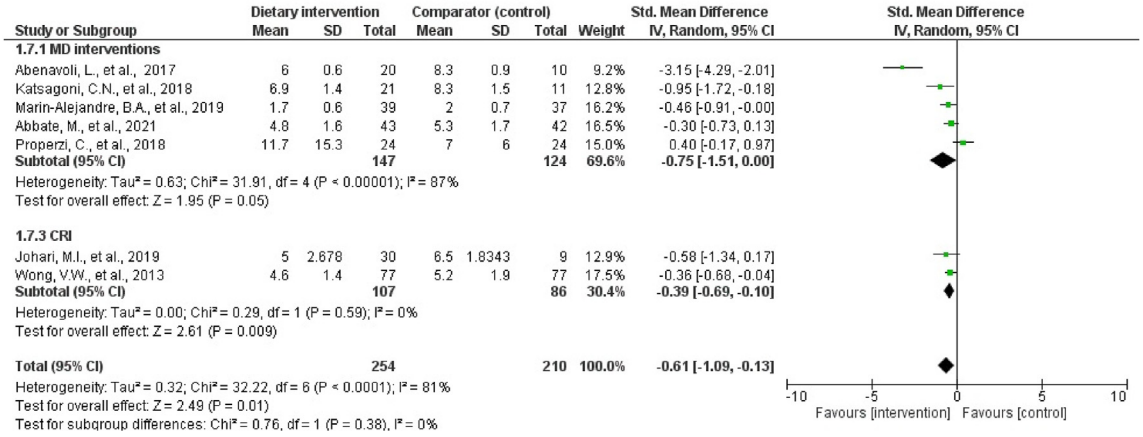

Fig. 8. Effects of dietary interventions on liver stiffness measurement (LSM). SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

4. Discussion

We identified, and analysed, data from 26 dietary intervention studies in adults with NAFLD that reported effects on markers of liver function, weight loss and on attrition during the intervention. Our systematic review and meta-analysis provides up-to-date evidence supporting the delivery of dietary interventions in the management of NAFLD. The evidence from the reviewed trials suggests a range of dietary interventions improve surrogate markers of NAFLD over periods of two weeks to two years. Our

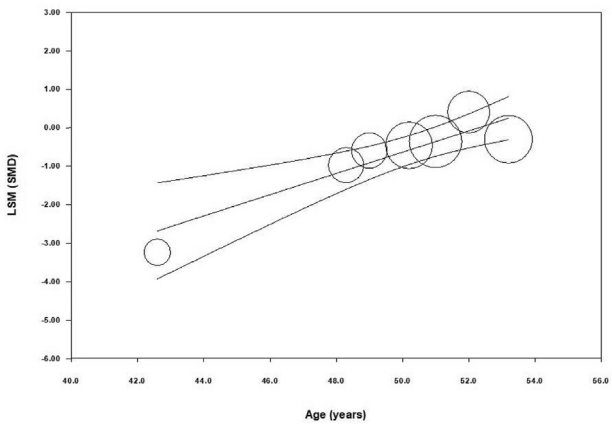

Fig. 9. Meta-regression of effects of age on liver stiffness measurement (LSM) change after dietary intervention. SMD, standardised mean difference.

meta-regression results suggest that diet-induced improvement in liver stiffness was greatest in the youngest patients and decreased with age. The results of the sensitivity analyses did not greatly influence the overall findings and conclusions of the present review. There remains uncertainty about the effects of lifestyle interventions on clinical outcomes, such as mortality, hepatic cirrhosis, and HCC, which are unlikely to be noted in trials with less than five to ten years follow-up [54].

4.1. The effectiveness of calorie-restricted interventions

Across the reviewed trials, calorie-restricted interventions produced improvements in liver function. The effectiveness of calorie restriction on ALT, HS and LSM shown in our meta-analysis findings, suggests this strategy should remain the cornerstone of NAFLD management. The reviewed trials used a range of specific dietary treatments or broader healthy dietary guidance, with or without general physical activity advice, to induce from modest to more severe calorie restriction.

There was limited evidence that over 2-weeks, a carbohydrate-restricted diet reduced liver fat more than a calorie-restricted diet [48]. Currently, there is no clear definition of a low carbohydrate diet, though <20 g carbohydrate/d used in this trial could be regarded as a very low carbohydrate, potentially ketogenic, diet [55]. There are few data on the adverse effects of these type of diets, and of their long-term outcomes in people with NAFLD.

Reduction in body weight of ~5% in NAFLD and ≥7% in NASH appeared beneficial, and achievable. Two of the longest and most intensive multi-component interventions induced weight loss ≥7% [46,51]. These interventions targeted sustained weight loss, delivered nutritional/behavioural counselling and advised ~200 min/week of moderate intensity physical activity, with or without resistance training. However, the dietary components of the two interventions were quite different. A community intervention advised calorie reduction with an individualized menu plan over 12-months [46]. The other intervention (lasting 48 weeks) used a more severe calorie restriction of 1000–1500 cal/d with commercial portion-controlled foods, shifting to broader dietary guidance over time [51]. The more hypocaloric approach followed by flexible dietary restraint resulted in slightly higher weight loss (9.3% vs. 8.0% for the community intervention) [46,51].

Limited evidence showed the advantageous effects of both calorie-restriction and dietary change components on cardiometabolic parameters [32,46,50,52] and abdominal obesity [46]. This may have important clinical implications for risk reduction of CVD and T2DM in patients with NAFLD. The modest average attrition rates suggests that the interventions were broadly acceptable to the participants. Moreover, limited data showed that the interventions induced diet intake modification as prescribed.

The current evidence shows mixed findings from a limited number of studies and further clarification from experimental trials is needed to determine the dietary components associated with the

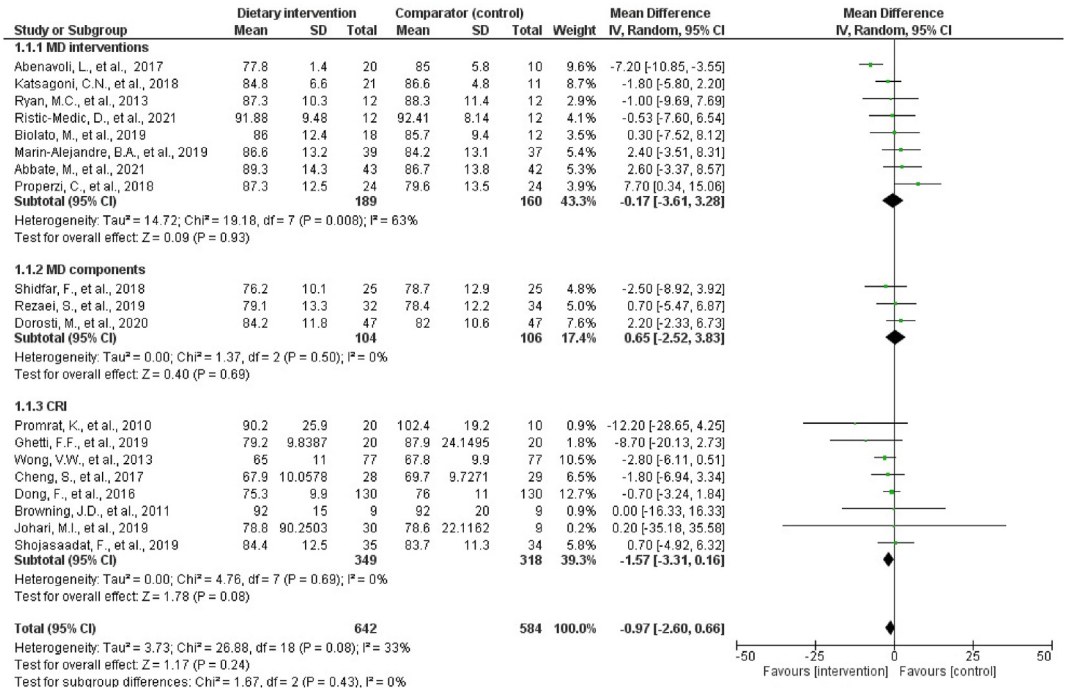

Fig. 10. Effects of dietary interventions on body weight. SD, standard deviation; IV, inverse variance; CI, confidence interval; MD, Mediterranean diet; CRI, calorie-restricted interventions.

greatest benefits on liver-related outcomes. In the context of moderate weight loss, a small number of trials suggested that specific dietary components, macronutrients and dietary patterns might improve liver function in those with NAFLD. Because recommended weight loss targets are difficult to achieve and to sustain, better evidence of the efficacy of different diet composition modifications could be a basis for developing personalised dietary care.

4.2. The effectiveness of MD and MD component interventions

There was consistent evidence that MD interventions improved one or more measures of NAFLD activity and meta-analysis showed a significant effect on post-intervention ALT, FLI and LSM. This is consistent with recent reviews which concluded that MD interventions improved FLI [56], liver enzymes and NAFLD severity indices [57]. Very limited evidence suggested that individual components of MD such as wholegrains and olive oil may improve NAFLD surrogate markers [43–45], but meta-analysis revealed no significant effects. Since the reviewed trials adopted different variants of MD, the exact dietary pattern associated with the benefits is unclear and the mechanisms through which MD exerts its NAFLD-specific effects remain poorly understood.

The MD pattern, without calorie restriction, appeared to exert advantageous effects even with minimal weight loss (~2%) [34,37] which suggest that MD patterns may have potential to improve NAFLD with or without weight loss. However, calorie-restricted MD interventions resulted in weight loss ranging from >5% [27,35,38,40] to 7–10% [33,39,42], which is in line with the European Association for the Study of the Liver (EASL) clinical practice guidelines [8]. Thus, the combination of calorie restriction and MD may be an achievable strategy to induce the recommended degree of weight loss and to promote improvements in NAFLD.

Limited evidence showed that MD interventions produced favourable changes in cardiometabolic measures [29,33,38] and abdominal obesity [28,29,38,40]. Protective effects of MD on MetS features, which are strongly associated with NAFLD, have been reported [58]. This is in line with previous reviews, which found that MD improved insulin resistance [56], lipid profile, and glycaemic indices [57]. The slightly lower attrition rates (10% in MD groups vs. 14% in comparator (control) groups (excluding the trial with MD in both groups) [27]), supports the idea that MD is an acceptable diet therapy in this patient population. The reviewed trials suggest that, over periods of eight weeks to two years, the interventions were successful in increasing MD adherence or altering intake of its components.

Current evidence of MD effectiveness comes mainly from trials conducted in Mediterranean regions/climates with favourable food environments and cultural dietary habits. Thus, any potential impact on disease outcomes may be affected by baseline MD adherence [59]. This may affect the potential transferability of findings in populations with different habitual dietary patterns. However, previous research has shown that short-term interventions can increase MD adoption and maintenance for up to one year in non-Mediterranean countries [16,60–62].

5. Characteristics of effective interventions

A key objective when evaluating diet lifestyle interventions is to determine whether interventions are implementable if shown to be effective [59]. Moreover, with increasing interest in personalised nutrition, the aim is to identify individuals or subgroups who may benefit from a specific intervention. This is challenging in the present review, given that the interventions ranged from simple to more complex with heterogeneity in delivery mechanisms, intensity and behavioural strategies.

Diet intake assessment typically relied on self-report instruments, each of which has well-described risks of misreporting, errors and bias. The most common imaging tool was MR, which is the most accurate method among imaging surrogates for the assessment of HS [63], but this was used by less than half of trials. This was followed by USnd, while this is a practical tool, it has limitations [64]. The FLI and NFS were the most commonly-used validated, non-invasive scoring systems in the reviewed trials. LSM was reported in less than a third of trials and performed by various methods. While invasive and subject to sampling variability, liver biopsy is the current gold standard for NAFLD assessment, but histologic examination was conducted in only one trial [51]. The need for accurate non-invasive assessment of NAFLD is yet to be addressed.

The main intervention delivery mechanism was face-to-face in individual consultations with smaller numbers of studies using group sessions and telephone contacts. A web-based intervention that included some face-to-face interaction was as effective as a standard group intervention in NAFLD [27]. Although internet approaches may reduce burden on patients and healthcare systems, a higher attrition rate was reported in the web-treated group [27]. Successful dietary change intervention requires strong participant adherence [12,16] and more evidence of the impact of dietary interventions on patient-related outcomes such as quality of life, dietary adherence and acceptability is needed.

The choice of intervention setting, and the involvement of healthcare professionals, may influence the adoption/maintenance of dietary changes. The current evidence came mainly from specialist/general settings, with fewer trials conducted in the community [28,29,46]. Nutritionists and dietitians were responsible for most of the interventions targeting dietary components. Although, some of the reviewed trials utilised personalised approaches to improve intervention adherence and effectiveness, more evidence of the most important individual characteristics for use in such personalised approaches would advance the development of interventions tailored to different NAFLD subgroups in different settings.

Less than a quarter of trials described applying specific behaviour change strategies, including stimulus control [27,51], goal setting [35], motivational interviewing [27], coping with risky situations [46], problem solving and relapse prevention [51] and weight loss maintenance strategies [27]. A lifestyle modification intervention incorporated the health belief model (HBM), improving knowledge, HBM variables and NAFLD surrogate markers [41]. However, which specific behaviour change strategies enhance intervention effectiveness remains uncertain. This should be a priority for future research to optimise intervention delivery, induce sustained dietary behaviour change and maximise benefits on liver function.

6. Conclusion

The current evidence suggests that most dietary interventions investigated in these trials were acceptable, and their adoption was associated with positive clinical outcomes. Calorie restriction, delivered via a range of interventions, is effective with a dose–response association between weight loss and effects on NAFLD activity. Changing diet composition has potential to complement diet-induced weight loss as the main therapy for NAFLD, and the limited data suggest that MD may be an effective treatment. However, further information from intervention trials that use individual participant characteristics to develop targeted diet therapies, would be advantageous.

There remain uncertainties about what is the most effective dietary intervention for patients with NAFLD. The substantial

heterogeneity in the reviewed studies, and the limited direct inter-trial comparisons restrict potential transferability of findings into clinical practice. Trials were short to medium term and findings need to be verified in larger long-term studies using consistent outcomes measures. There was also limited evidence on the factors influencing individual patient decisions to adopt, and capacity to maintain, dietary changes. Key recommendations for future studies include robust assessment of dietary intake, intervention acceptability and sustainability, and quality of life and patient-related outcomes. This information may guide personalised nutrition intervention approaches and improve care of patients with NAFLD.

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Author contributions

The authors' contributions were as follows - LH, JCM, and QMA conceptualised the review and formulated the systematic search strategy; LE assisted with the formulation of search strategies; LH conducted the database searches; LH, CK, JG and JCM assessed article eligibility and risk of bias; LH and CK performed data extraction; LH conducted meta-analysis and meta-regression; KE assisted with meta-analysis and meta-regression; LH drafted the manuscript; and all authors read and approved the final manuscript.

Conflict of interest

No conflicts of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnu.2022.06.037.

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